Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To learn the Basics of Machine Learning**	
Experiment No: 01	**Date:**	**Enrollment No: 92410133038**

Aim: To learn the Basics of Machine Learning

IDE: Google Colab

Theory:

Machine learning is a subset of artificial intelligence in the field of computer science that often uses statistical techniques to give computers the ability to "learn" (i.e., progressively improve performance on a specific task) with data, without being explicitly programmed. In the past decade, machine learning has given us self-driving cars, practical speech recognition, effective web search, and a vastly improved understanding of the human genome.

Pre Lab Exercise:

A. Write 5 applications, that you encounter daily, where Machine Learning is used?

1. YouTube Recommendations
2. Instagram Feed
3. Email Spam Filter
4. Google maps navigation
5. Google Search Engine

B. What do you mean by Supervised Learning?

It is a ML method where the model trained using the label Data . Supervised learning is a type of machine learning in which the computer is provided with example inputs along with their desired outputs, given by a “teacher”. The goal of the system is to learn a general rule that maps the inputs to the outputs.

C. What do you mean by Unsupervised Learning?

It is type of ML in which algorithm finds patterns and structures in data without Labled outputs. Unsupervised learning is a type of machine learning where no labels are given to the learning algorithm. The system is not provided with the correct outputs. It is left on its own to find structure or patterns in the input data

D. What do you mean by Reinforcement Learning?

Here, agent will learn by trial and error to maximize a reward

It is a Experimental learning

Reinforcement learning is a type of machine learning in which an agent learns by interacting with an environment. The training data is given in the form of rewards and punishments based on the actions taken by the program. The main goal of reinforcement learning is to find a policy that helps the agent take actions in such a way that it maximizes long-term reward. Unlike supervised learning, correct input and output pairs are not given directly.

Program (Code):

Perform the following tasks:

1. Load a dataset in your IDE
2. Observe the statistics of all the features
3. Obtain the shape of the dataset
4. Separate all the features
5. Fill the missing values, if any, using the statistically relevant value
6. Observe the Box-Plot of each feature
7. Comment on the IQR range and outliers

Results:

Code:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

df = pd.read_csv("sample_data/california_housing_train.csv")

print("Statistical Summary:")

print(df.describe())
```

After Run, Cell Output

Statistical Summary:

```
         longitude      latitude  housing_median_age   total_rooms  \
count  17000.000000  17000.000000        17000.000000  17000.000000
mean    -119.562108     35.625225           28.589353   2643.664412
std        2.005166      2.137340           12.586937   2179.947071
min     -124.350000     32.540000            1.000000      2.000000
25%     -121.790000     33.930000           18.000000   1462.000000
50%     -118.490000     34.250000           29.000000   2127.000000
75%     -118.000000     37.720000           37.000000   3151.250000
max     -114.310000     41.950000           52.000000  37937.000000
       total_bedrooms    population    households  median_income  \
count    17000.000000  17000.000000  17000.000000   17000.000000
mean       539.410824   1429.573941    501.221941       3.883578
std        421.499452   1147.852959    384.520841       1.908157
min          1.000000      3.000000      1.000000       0.499900
25%        297.000000    790.000000    282.000000       2.566375
50%        434.000000   1167.000000    409.000000       3.544600
75%        648.250000   1721.000000    605.250000       4.767000
max       6445.000000  35682.000000   6082.000000      15.000100

       median_house_value
count        17000.000000
```

```
mean        207300.912353
std         115983.764387
min          14999.000000
25%         119400.000000
50%         180400.000000
75%         265000.000000
max         500001.000000
```

```
print("Shape of dataset (rows, columns):", df.shape)
```

After run, Cell Output

```
Shape of dataset (rows, columns): (17000, 9)
```

```
X = df.iloc[:, :-1]   # Independent features
Y = df.iloc[:, -1]    # Target feature
print("Independent Features:")
print(X.head())
```

After Run, Cell Output

```
Independent Features:
   longitude  latitude  housing_median_age  total_rooms  total_bedrooms  \
0    -114.31     34.19                15.0       5612.0          1283.0
1    -114.47     34.40                19.0       7650.0          1901.0
2    -114.56     33.69                17.0        720.0           174.0
```

```
3   -114.57   33.64        14.0   1501.0      337.0
4   -114.57   33.57        20.0   1454.0      326.0

   population  households  median_income
0      1015.0       472.0         1.4936
1      1129.0       463.0         1.8200
2       333.0       117.0         1.6509
3       515.0       226.0         3.1917
4       624.0       262.0         1.9250
```

```
print("Target Feature:")
print(Y.head())
```

After Run, Cell Output

```
Target Feature:
0    66900.0
1    80100.0
2    85700.0
3    73400.0
4    65500.0
Name: median_house_value, dtype: float64
```

```
print("Missing values before filling:")
print(df.isnull().sum())
```

After Run, Cell Output

```
Missing values before filling:
longitude            0
latitude             0
housing_median_age   0
total_rooms          0
total_bedrooms       0
population           0
households           0
median_income        0
median_house_value   0
dtype: int64
```

```
# Filling missing numerical values with mean
df.fillna(df.mean(numeric_only=True), inplace=True)
print("Missing values after filling:")
print(df.isnull().sum())
```

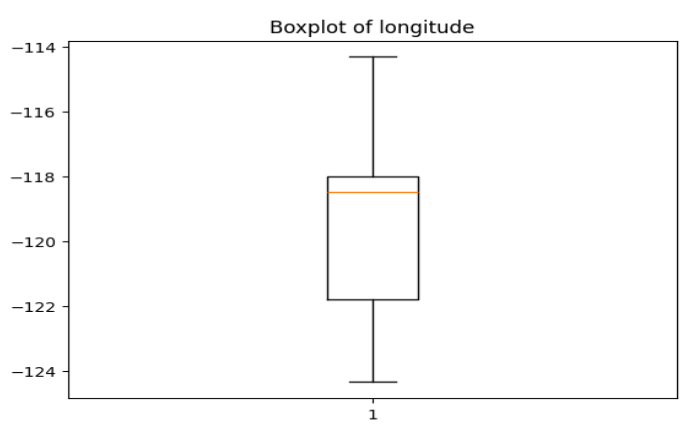
After Run, Cell Output

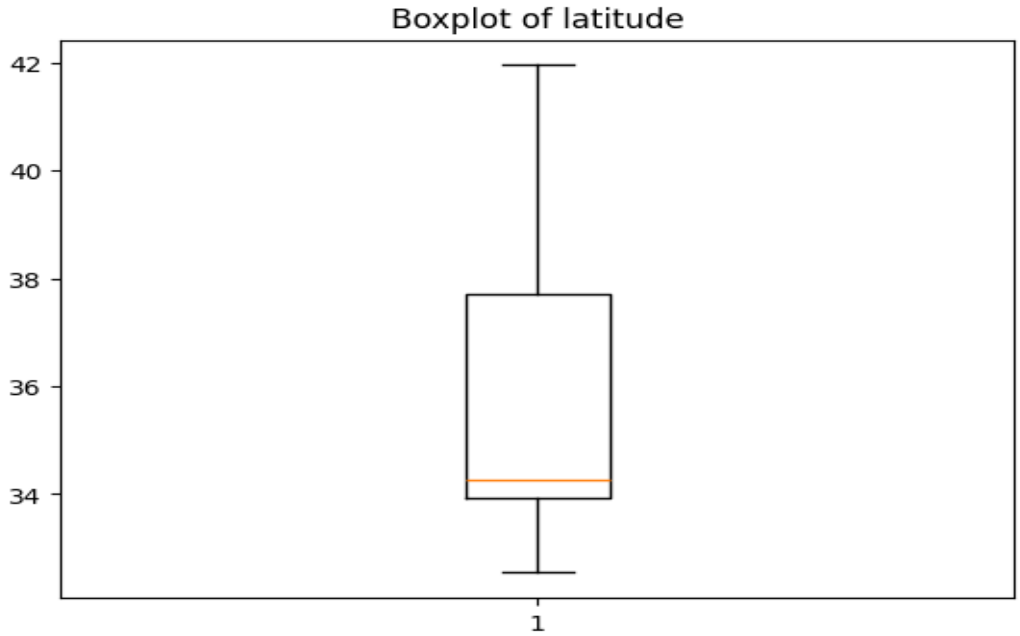
```
Missing values after filling:
longitude            0
```

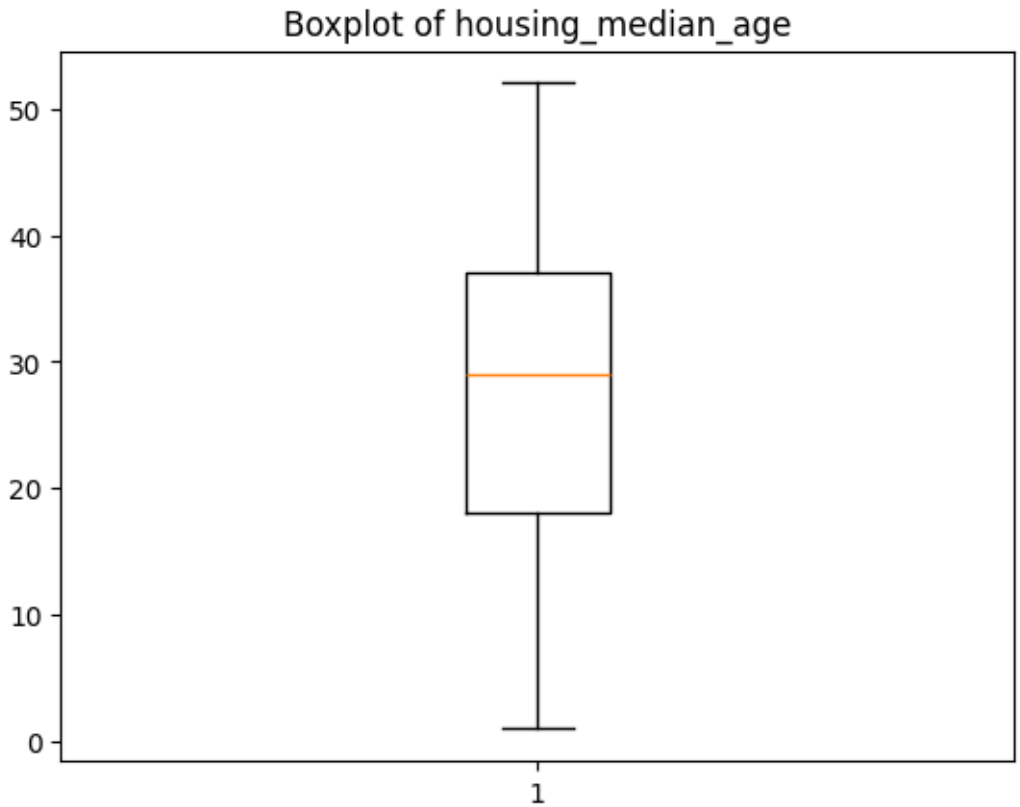
```
latitude            0
housing_median_age    0
total_rooms          0
total_bedrooms        0
population           0
households           0
median_income        0
median_house_value    0
dtype: int64
```

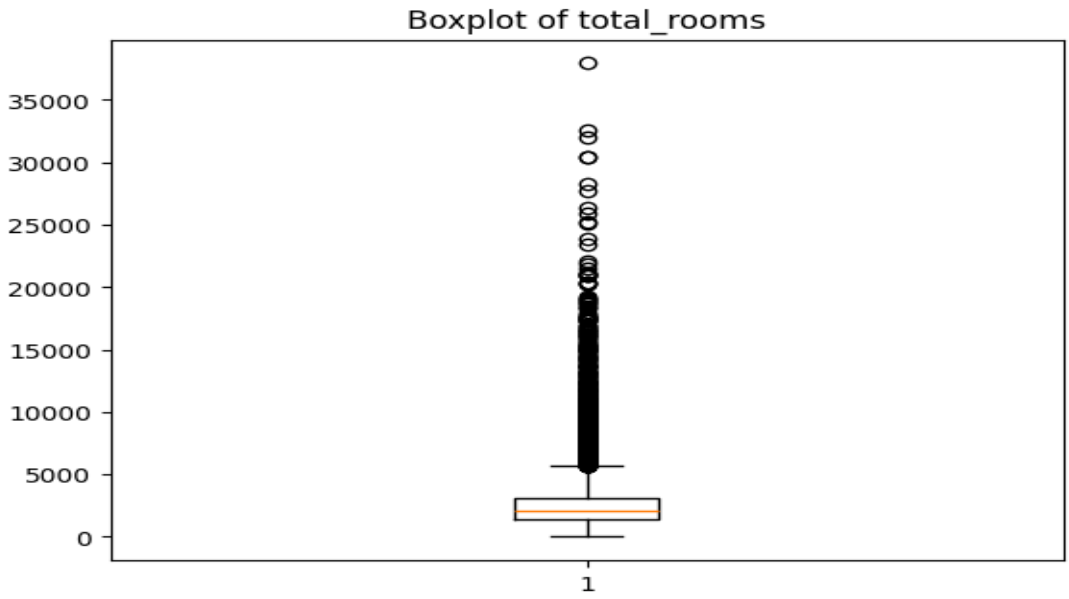
```
for column in df.select_dtypes(include=np.number).columns:
    plt.figure()
    plt.boxplot(df[column])
    plt.title("Boxplot of " + column)
    plt.show()
```

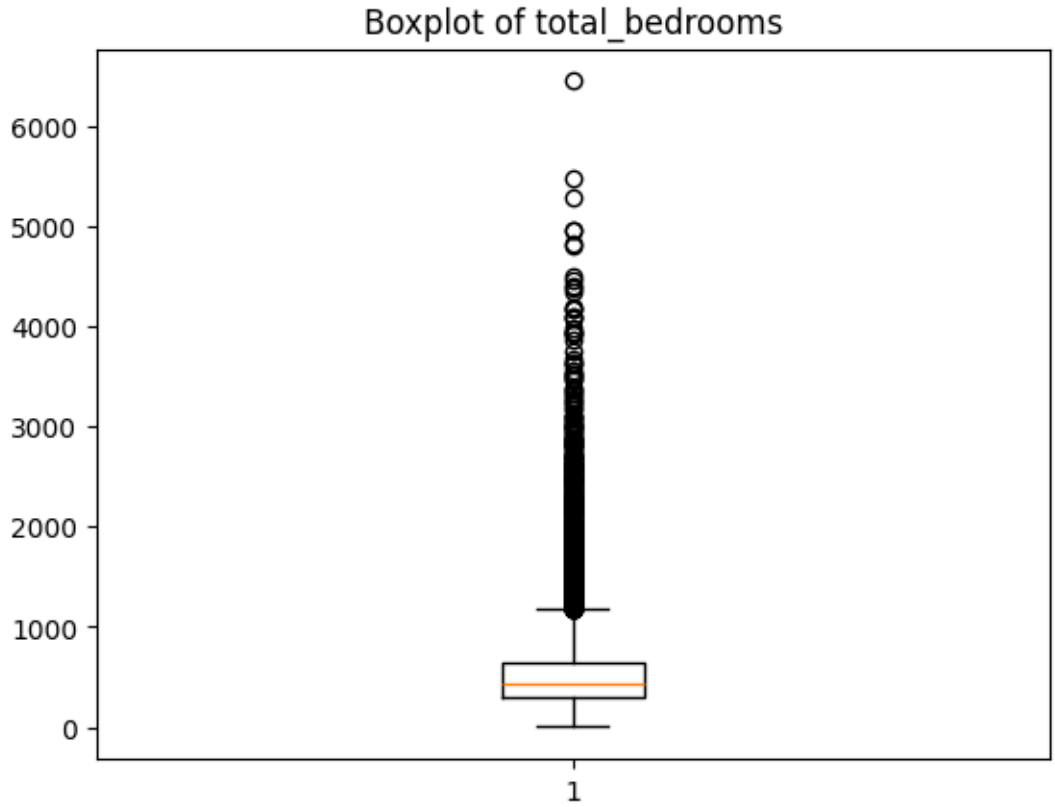
After Run, Cell Output

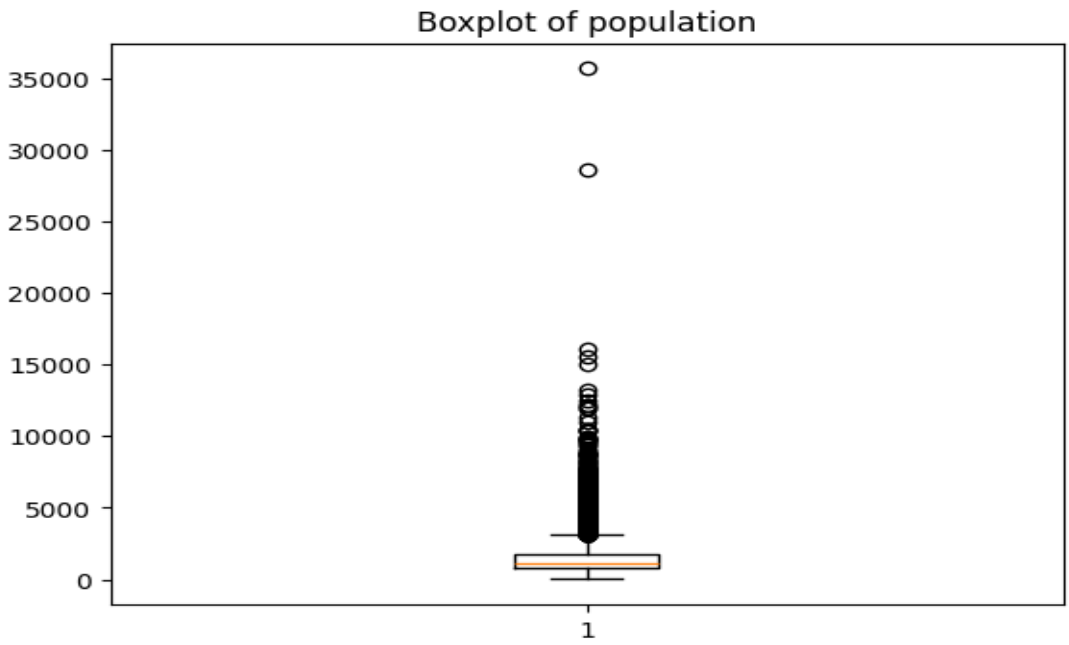


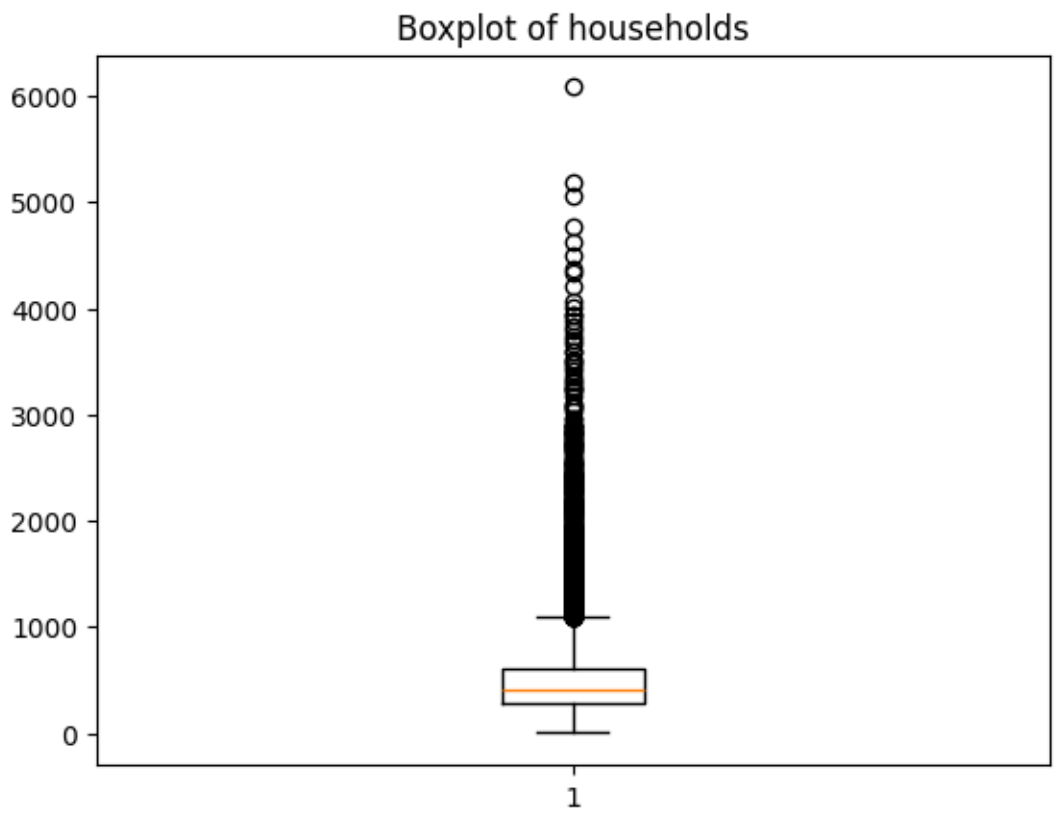

Boxplot of latitude
42
40
38
36
34
1

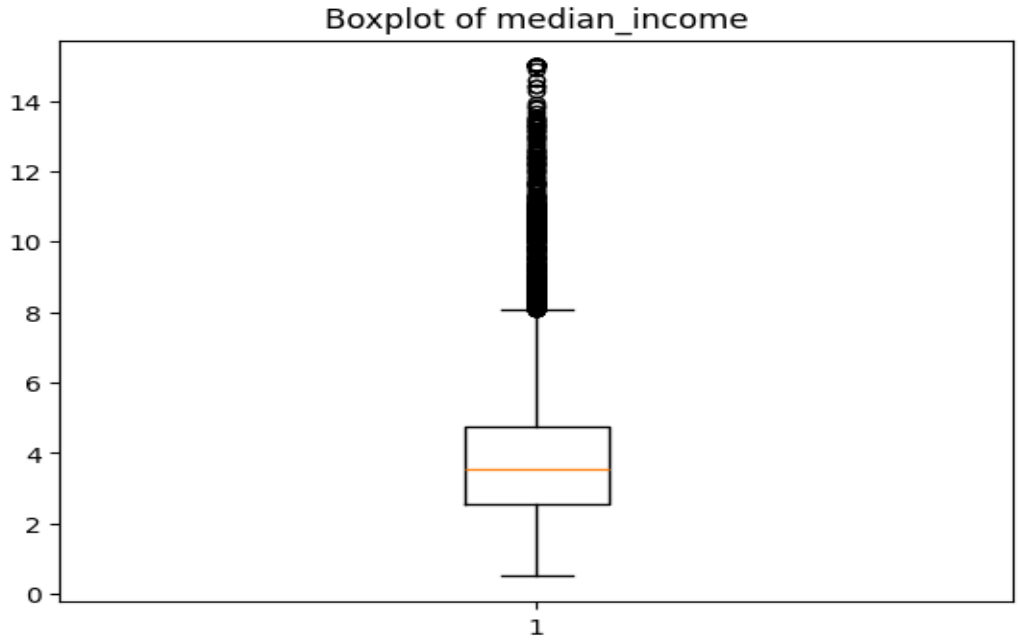

Boxplot of housing_median_age
50
40
30
20
10
0
1

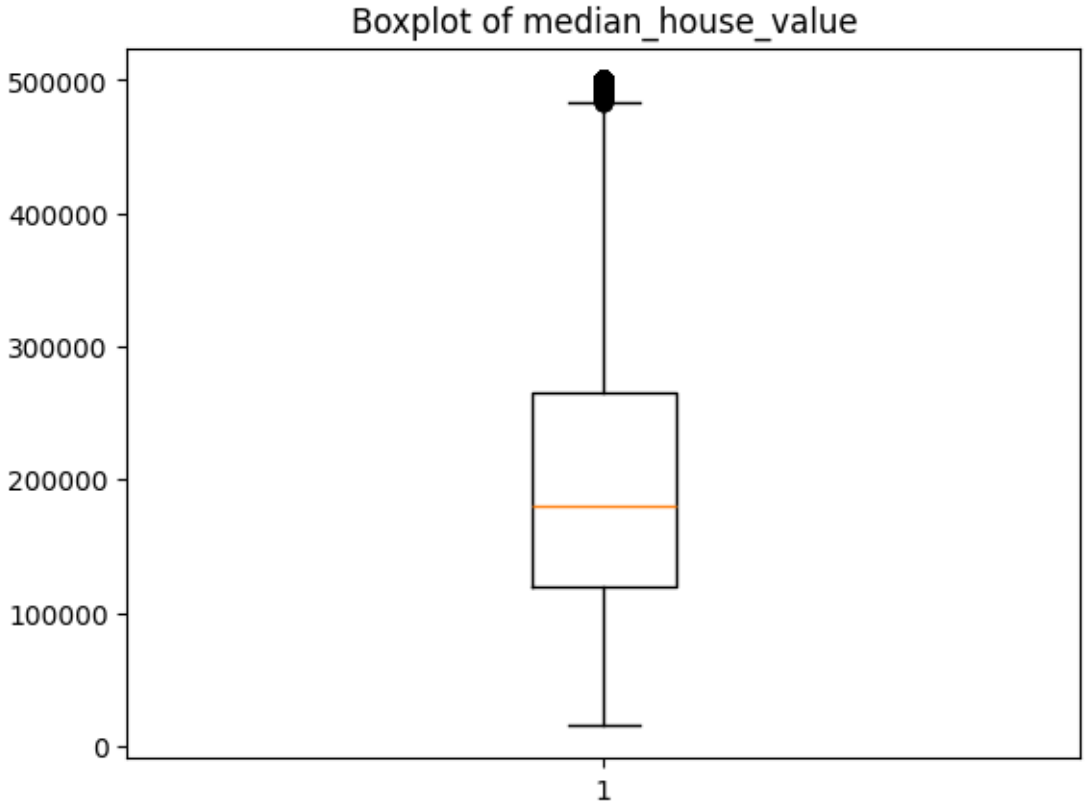

Boxplot of total_rooms
35000
30000
25000
20000
15000
10000
5000
0
1


Boxplot of total_bedrooms
6000
5000
4000
3000
2000
1000
0
1


Boxplot of population
35000
30000
25000
20000
15000
10000
5000
0
1


Boxplot of households
6000
5000
4000
3000
2000
1000
0
1


Boxplot of median_income
14
12
10
8
6
4
2
0
1


Boxplot of median_house_value
500000
400000
300000
200000
100000
0
1

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To learn the Basics of Machine Learning**	
Experiment No: 01	**Date:**	**Enrollment No: 92410133038**

RESULT:- (Codes & Outputs):-

colab.research.google.com/drive/1D5pjQzD4IixZKMOoGtiLxBKEQfUMekmn#scrollTo=M70ORg-JBdG6

Lab_01

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[2]
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

[4]
```python
df = pd.read_csv("sample_data/california_housing_train.csv")
```

Lab_01

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
Statistical Summary:
          longitude      latitude  housing_median_age   total_rooms  \
count  17000.000000  17000.000000        17000.000000  17000.000000
mean    -119.562108     35.625225           28.589353   2643.664412
std        2.005166      2.137340           12.586937   2179.947071
min     -124.350000     32.540000            1.000000      2.000000
25%     -121.790000     33.930000           18.000000   1462.000000
50%     -118.490000     34.250000           29.000000   2127.000000
75%     -118.000000     37.720000           37.000000   3151.250000
max     -114.310000     41.950000           52.000000  37937.000000

       total_bedrooms    population    households  median_income  \
count    17000.000000  17000.000000  17000.000000   17000.000000
mean       539.410824   1429.573941    501.221941       3.883578
std        421.499452   1147.852959    384.520841       1.908157
min          1.000000      3.000000      1.000000       0.499900
25%        297.000000    790.000000    282.000000       2.566375
50%        434.000000   1167.000000    409.000000       3.544600
75%        648.250000   1721.000000    605.250000       4.767000
max       6445.000000  35682.000000   6082.000000      15.000100

       median_house_value
count       17000.000000
mean       207300.912353
std        115983.764387
min         14999.000000
25%        119400.000000
50%        180400.000000
75%        265000.000000
max        500001.000000
```

[7]
```python
print("Shape of dataset (rows, columns):", df.shape)
```

```
Shape of dataset (rows, columns): (17000, 9)
```

[8]
```python
X = df.iloc[:, :-1]   # Independent features
Y = df.iloc[:, -1]    # Target feature
```

[10]
```python
print("Independent Features:")
print(X.head())
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To learn the Basics of Machine Learning**	
Experiment No: 01	**Date:**	**Enrollment No: 92410133038**

```
Independent Features:
   longitude  latitude  housing_median_age  total_rooms  total_bedrooms  \
0    -114.31     34.19                15.0       5612.0          1283.0
1    -114.47     34.40                19.0       7650.0          1901.0
2    -114.56     33.69                17.0        720.0           174.0
3    -114.57     33.64                14.0       1501.0           337.0
4    -114.57     33.57                20.0       1454.0           326.0

   population  households  median_income
0      1015.0       472.0         1.4936
1      1129.0       463.0         1.8200
2       333.0       117.0         1.6509
3       515.0       226.0         3.1917
4       624.0       262.0         1.9250
```

colab.research.google.com/drive/1D5pjQzD4IixZKMOoGtiLxBKEQfUMekmn#scrollTo=M70ORg-JBdG6

Lab_01

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[11]

```
print("Target Feature:")
print(Y.head())
```

```
Target Feature:
0    66900.0
1    80100.0
2    85700.0
3    73400.0
4    65500.0
Name: median_house_value, dtype: float64
```

[12]

```
print("Missing values before filling:")
print(df.isnull().sum())
```

```
Missing values before filling:
longitude             0
latitude              0
housing_median_age    0
total_rooms           0
total_bedrooms        0
population            0
households            0
median_income         0
median_house_value    0
dtype: int64
```

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To learn the Basics of Machine Learning**	
Experiment No: 01	**Date:**	**Enrollment No: 92410133038**

```
[13]
✓ 0s
# Filling missing numerical values with mean
df.fillna(df.mean(numeric_only=True), inplace=True)
print("Missing values after filling:")
print(df.isnull().sum())

Missing values after filling:
longitude             0
latitude              0
housing_median_age    0
total_rooms           0
total_bedrooms        0
population            0
households            0
median_income         0
median_house_value    0
dtype: int64

[14]
✓ 1s
for column in df.select_dtypes(include=np.number).columns:
    plt.figure()
    plt.boxplot(df[column])
    plt.title("Boxplot of " + column)
    plt.show()
```

Observation and Result Analysis:

A. Statistics of the dataset

From the statistical summary obtained using the describe() function, we observed the count, mean, standard deviation, minimum value, maximum value, and quartile values (25%, 50%, and 75%) for all numerical features.

The count value shows that most of the features have equal number of records, which means there are very few or no missing values. The mean and median (50%) values help in understanding the central tendency of the data. If the mean and median values are close, the data is normally distributed.

The standard deviation indicates how much the data is spread from the mean. Features with higher standard deviation show more variation in data. The minimum and maximum values give the range of the data.

Overall, the statistical summary helps to understand the distribution, spread, and nature of the dataset before further processing.

B. IQR and Outliers

From the boxplot of each numerical feature, the Inter Quartile Range (IQR) was observed. The IQR is calculated as the difference between the third quartile (Q3) and the first quartile (Q1), i.e., IQR = Q3 – Q1. It represents the middle 50% of the data.
Using the IQR method, values below Q1 – 1.5×IQR or above Q3 + 1.5×IQR are considered outliers. In the boxplot, these outliers are shown as points outside the whiskers.
From the observed boxplots, some features showed very few outliers, while some features had no significant outliers. The IQR range helped in understanding the spread and variability of the data. Features with larger IQR indicate more spread in the middle portion of the dataset.

<u>Post Lab Exercise:</u>

A. Which is a better mean of replacing the missing value-mean, median or mode?

The better method of replacing missing values depends on the type and distribution of data. For numerical data, mean or median can be used. If the data does not contain outliers and is normally distributed, mean is suitable. If the data contains outliers, median is better because it is not affected much by extreme values.

For categorical data, mode is generally used because it replaces the missing value with the most frequently occurring category.

B. How IQR range helps to understand the nature of the distribution of the data?

The IQR (Inter Quartile Range) represents the difference between Q3 and Q1 and shows the spread of the middle 50% of the data. It helps to understand how the data is distributed and whether the data is spread out or closely packed. IQR is also useful in detecting outliers. Values lying below Q1 – 1.5×IQR or above Q3 + 1.5×IQR are considered outliers. Therefore, IQR helps in understanding the variability and the presence of extreme values in the dataset.

C. Which are the ways to replace the categorical data to the numerical ones?

Categorical data can be converted into numerical form using different encoding techniques. One common method is assigning numerical labels to each category. Another method is creating separate binary columns for each category (dummy variables).

These methods help in converting categorical data into numerical form so that machine learning algorithms can process the data properly.

Conclusion:

In this experiment, the basic concepts of Machine Learning were studied and implemented using Google Collab. The dataset was successfully loaded and analyzed. By observing its statistical summary and shape. Missing values were identified and handled using appropriate statistical methods. Box plots were also generated to visualize the distribution of features and to detect possible outliers using the IQR method. Overall, this experiment helped in understanding the initial steps involved in data analysis and preprocessing before applying machine learning models.